THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

THE GEMINI PROTOCOLS · TOPICAL BOOK — LIVING VOLUME

THE BATHROOM BOOK

FOR:

The people who will actually plumb it.

ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPhD

COVER ISSUED 21 AUGUST 2026 — GEMINI 1 × KIMI CHAT 3

THE BATHROOM BOOK — EXACT BUILDER'S INSTRUCTIONS

Ship Sanitation Suite: Phases 137 (toilet), 146 (shower), 145 (the water rule), 136 (bottle), 144 (dry-waste chute)

Co-Authors: Archer Quinn, Gemini 1 & Kimi Chat (The Initial Interstellar Co-Engineering Alliance)

What this book is: the bathroom phases of the Gemini Protocols rewritten as exact, numbered build instructions in plain English — no markup language, no cross-references you have to chase. Read the rule in section one; everything after it is just plumbing that obeys.

SECTION 1 — THE ONE RULE EVERYTHING OBEYS (PHASE 145)

On Earth, water falls down. In space there is no down. A drop of water leaving a nozzle travels in a straight line, at the speed you gave it, until it hits something. That is the entire physics of this bathroom:

- **Rule 1:** Water goes where you throw it. Every nozzle is an aiming device, not a tap.
- **Rule 2:** Water clings to skin (surface tension). Left alone it sheets over a face and stays there — this is the drowning hazard, and it is why astronauts normally wash with wipes. This bathroom beats it with moving air and cold plates, not gravity.
- **Rule 3:** Spinning makes 'down' for free. Anything you rotate flings its contents outward to the wall — that fake gravity is what the toilet and the floor drains run on.
- **Rule 4:** Cold pulls. Water vapor drifts from warm zones to cold surfaces and condenses there. The shower's whole exhaust system is one cold wall. No fans screaming, no hoses sucking.

If a builder remembers nothing else: aim it, spin it, or freeze it. Those three verbs replace gravity.

SECTION 2 — THE TOILET (PHASE 137): THE SPINNING BOWL

What it is: a motorized polycarbonate bowl that spins under an airtight seat. Fake gravity flings everything outward into a collection gutter. No vacuum suction, no screaming fans, no floating mistakes.

Build it in this order:

- **1. The seat frame.** Static (non-rotating), airtight against the bowl rim. The seal does the work of a bathroom door lock — nothing escapes past it.
- **2. The bowl.** Polycarbonate, spherical-cone profile, motor-driven from below. Spins fast enough to pin liquid against its walls (the same rule as a salad spinner).

- **3. Water injection — the prewet.** Spray grids prewet the bowl walls the moment you sit (provenance logged: the audit's suggestion, adopted by the author — Phase 137 Amendment 2). The moving sheet of water is the cleaning cycle and the transport medium in one.
- **4. The collection gutter — OPEN FACE, TRIANGLE BACK (corrected 5 August 2026, Phase 137 Amendments 1-3).** A vertical gutter in the side wall of the bowl, running top to bottom, its face OPEN to the bowl interior — no lip, no rim: the spinning wall carries everything around and the open face simply stands in its path, so collection is automatic and nothing ever falls. The gutter's rear goes TRIANGLE: the channel narrows to a point at its back, so fines wedge and flush instead of packing, and the prewet sheet sweeps the point clean. Spin pressure drives the collected mass along the gutter to the exit pipe.
- **5. The flap valve.** A 3-inch spring-loaded flap on the exit line. It stays snapped shut; only the pressure of actively arriving spun mass opens it. When the spin stops, the line seals itself.
- **5a. The swivel pipe (Amendment 3).** The gutter's exit pipe is hard-mounted against the bowl — snug, not loose, nothing flapping — and it CO-ROTATES with the bowl. One swivel joint (a rotary union) makes the only moving-to-static handoff, passing spun mass to the static transit line under its own spin load. All the wear lives in that one swivel; it's a replaceable cartridge.
- **6. The transit line.** An externally-driven rifled conduit (Phase 129) — the pipe's wall rotates via an outside helical track, so there is no internal auger shaft to clog. Waste rides the moving wall away.
- **7. The cartridges.** High-use conduit sections are swappable polycarbonate cartridges on toggle clips. Pull a dirty one, shred and purify it, re-print it into a new pipe. The plumbing recycles itself.

Earth test: mount the bowl on a variable-speed turntable, pour in dyed water, and watch it climb the walls into the gutter. If the gutter catches everything at spin-up, the physics is proven before lunch.

SECTION 3 — THE SHOWER (PHASE 146): THE SNUB-NOSED BULLET

What it is: a shower stall shaped like a snub-nosed bullet — warm high-pressure air at the front dome, a freezing cold wall at the rear. Water and steam drift to the cold side by nature, condense, and run into floor gutters. Zero suction machinery.

Amended 5 August 2026: the keyhole flare after the standing area runs deeper (a bigger diffuser — slower air, harder droplet drop-out, the high-low gradient maximised); the rear drain end shortens to about a foot beyond the cold plate — the offset maze already guarantees no line of sight; the perforated panel bank now runs full height to the ceiling with larger holes (offset kept — the maze works on offset, not smallness); and the air blade rides a vertical pole on its own hose, handheld like the shower head. One fitting language for water and air.

Build it in this order:

- **1. The shell.** Bullet profile: semicircular dome at the front, wide rectangular panels at the rear, person-sized.
- **2. The nose.** Mount handheld nozzles and low-speed warm-air plenums inside the front dome. This creates a gentle high-pressure zone where the user stands — air pushes from warm toward cold, and the water rides it.
- **3. The cold tail.** Link the rear panels to the Phase 96 cold sinks (or any refrigerated plate in testing). Thermal contraction draws every droplet and wisp of steam down the length of the stall to the back wall.
- **4. The colander maze.** The rear barrier is vertical panels spaced 1 inch apart, each perforated with 75mm holes at offset positions — no straight line of sight through. Air weaves through; water droplets cannot. They hit a panel and stay.
- **5. The freezing plate.** Behind the colander panels, one solid aluminum plate held sub-zero. Steam that reaches it condenses instantly — this is the final exhaust, and it is silent.
- **6. The floor.** Convex plates sloped to perimeter wall gutters (concave tracks per Phases 64/136). Condensed water runs the gutters to the recycling loop.
- **7. The air-knife.** A blade-fan slides up and down a vertical wall pole on foot pedals — hands stay free. It projects a razor-thin sheet of fast air that strips water off skin like a squeegee, always clearing the nose and mouth first. Detach it for a final handheld body sweep — 95% dry before you step out.

Earth test: in 1G this works even better (gravity helps the floor). Build the stall, run a fridge-cold rear plate, and measure how much water reaches the colander versus the floor. The maze and the cold plate should account for nearly all of it.

SECTION 4 — THE SPACE BOTTLE (PHASE 136): THE GUTTER DRINKER

What it is: the bathroom's drinking adjunct and the crew's personal bottle — a double-walled container whose inner lining has a downward-sloping gutter channel. A magnetic stirrer in the base spins the drink; dense liquid flings outward into the gutter, air bubbles gather at the center, and the straw sits in the gutter where only liquid reaches.

- **1.** Double-wall the bottle; form the inner wall with one vertical gutter tapering to a straw port at the perimeter.
- **2.** Embed a magnetic mixing bar in the base cavity, driven non-contact from outside — spin to sort liquid from bubbles.
- **3.** Anchor the straw terminal inside the gutter pocket; fit a sports-class flip-neck latch that seals airtight when closed.

- **4.** Earth version: lay the gutter as a horizontal base ring — stops protein sediment caking. This variant is a commercial fitness product as-is.

SECTION 5 — THE DRY-WASTE CHUTE (PHASE 144): THE CAPSULE POST

Status: partial survival. Phase 144's full markup was lost with the buffer wipe — but its core survived inside Phase 137 (section three), the same way the $2 springs survived inside Phase 117. What is certain:

- **1.** Dry hygiene papers and wipes are compressed into cardboard or gelatin capsules — never loose.
- **2.** The dry stream never touches the wet bowl system — paper fiber would blind the water-reclamation loops.
- **3.** A plain pneumatic vacuum-mail tube (the old department-store system) shoots the capsules straight to the Phase 73 charcoal kilns at 600C — waste to resource with zero handling.

Re-instruction prompts for Archer (answer in any words, I rebuild the phase): capsule size and how they're formed (hand press? machine?); inlet location per bathroom module; tube diameter and routing rules; what stops a capsule jam; whether the kiln intake needs an airlock stage; anything about the gelatin vs cardboard choice.

SECTION 6 — HOW THE WHOLE ROOM CONNECTS

- Shower water → colander maze → freezing plate → floor gutters → recycling loop (Phase 70 parallel lines).
- Toilet waste → 3-inch gutter → flap valve → rifled conduit → processing. Bowl wash water → same gutters.
- Dry paper → capsules → pneumatic tube → Phase 73 kiln (600C charcoal).
- Bottle and hand-rinse water → same gutter tracks — every drain in the room is a concave channel, because in space you route water along paths, not downhill.

The room in one sentence: nothing in this bathroom falls, sucks, or screams — water is aimed, spun, frozen, and posted, and the only moving parts a crew member touches are a straw latch and a foot pedal.

SECTION 7 — THE LAUNDRY: THE DROP-OFF (162) AND THE WRINGER LINE (191)

Corrected by Archer, 4 August 2026: this section used to present the plunger-lid basket as the washing machine. That machine description was made obsolete on 30 July 2026 — the plunger-lid basket stands as the DROP-OFF and transport standard (still true, as he said: true for dropping off clothes), and the washing is done by Phase 191's conveyor wringer line. The obsolete washer presentation stands untouched in Phase 162 of the Master File, marked OBSOLETE — superseded, never deleted.

Run it in this order:

- **1. The drop-off basket (162).** A cylinder loaded at bench height: clothes in one end, flexible mesh circular cover flexed in over them. Carry to the laundry full. One part, three jobs: the basket is the load, the lid, and its own base is the transfer plunger.
- **2. The drop pod (162, as amended 31 July 2026).** A multiroom collection tank in the deck floor above the laundry — tall as a big round tank, with a sealed top door of its own. The transfer sequence is law: place basket, lock in place, main pod door opens, open basket door, plunger pushed, basket door closed, main door closes, plunger up, return to room. Every move is gravity-indifferent — the plunger does in zero-G what falling does in gravity.
- **3. The wash (191).** From the pod, loads feed the wringer line: clothes ride a conveyor between upper and lower rubber roller sets an inch apart laterally, the lower set spring-loaded to hold constant nip pressure on any thickness. The load oscillates — a metre one way, then back — and every pass through the nip is a wash cycle.
- **4. The pressure flush (191).** Wash water injects through perforated roller faces directly into the compressed fabric at the nip. With the nip sealing lateral escape, the water's path of least resistance runs through the clothes — fabric is a baffle, and pressure flushes the matrix clean. No agitation paddles, no gravity drain, no spin.
- **5. The extraction (191).** The same wringer nip is the extractor — the mangle lineage, a century and a half of physics that never relied on gravity. Clothes leave the line wrung, not spun.
- **6. The drying room (162, stands).** Hang in the hot room with warm air blowing down — the downdraft holds clothes vertical — and vent the humid exhaust to the greenhouses. The laundry waters the farm.

Earth test: a roller wringer from any camping store — or a heritage mangle — proves steps 3 to 5 in an afternoon; a plunger bucket proves the drop-off handling; the rest is plumbing you already built.

For the record: the machine died but the containers lived — the basket-and-plunger architecture became the Phase 167 universal plunger container, the fleet's 10L/10kg supply-chain standard, and now closes the loop as the laundry's own throat. Nothing was wasted; the laundry machine was the container program's parent.